

TITLE PAGE



- **Problem Statement ID:** 4
- **Problem Statement Title:** AI-Powered Behavioral Anomaly Detection for Cybersecurity
- **Theme:** Cyber Security / Artificial Intelligence
- **PS Category:** Software
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- **Student ID:** 23BCE1326 (Vellore Institute of Technology, Chennai)

SentinelAI – AI-Powered Behavioural Anomaly Detection & SOC Risk Intelligence

Detailed explanation of the proposed solution

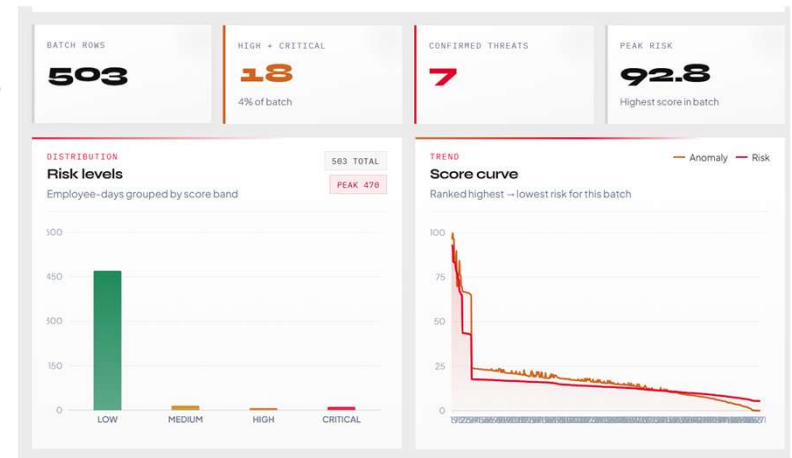
- AI-powered cybersecurity platform that learns normal user and device behaviour.
- Uses a Behavioural Transformer to detect anomalous activity without relying on attack signatures.
- Generates risk score, attack classification, MITRE ATT&CK mapping and explainable insights.
- Provides a complete SOC dashboard for monitoring, investigation and response.

How it addresses the problem

- Detects unknown and zero-day attacks through behavioural analysis.
- Identifies insider threats and credential misuse.
- Supports real-time monitoring and anomaly detection.
- Helps SOC analysts investigate incidents with explainable AI.

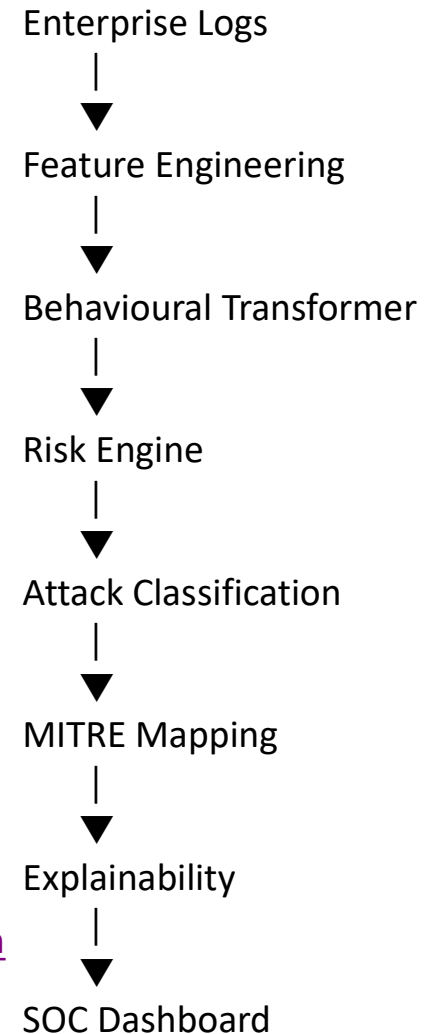
Innovation and uniqueness of the solution

- Behaviour-based detection instead of signature matching.
- Transformer-based anomaly detection with explainability.
- Handles cold-start users and concept drift.
- Multi-stage kill-chain correlation for campaign investigation.
- End-to-end cybersecurity platform from data generation to analyst dashboard.



TECHNICAL APPROACH

Layer	Technology / Stack
Data Processing	Python, Pandas, NumPy, Faker
Anomaly Detection	PyTorch Behavioural Transformer (Sequence Length: 64, Reconstruction + Attention)
Adaptation	Cold-Start Score Adjustment, Per-Entity EWMA Concept Drift Handling
Risk & Classification	Deterministic Risk Engine, Rule-Based Attack Classification, MITRE ATT&CK Mapping
Backend API	FastAPI, Pydantic, Uvicorn
SOC Dashboard	React 19, TypeScript, Vite, Recharts
Evaluation	Precision, Recall, F1-Score, Type Accuracy, FPR @ Top 1% & 5% (Simulator Ground Truth)
Artifact	Link
Live Demo	https://sentinel-ai-behavioral-threat-detection-keedprxpu.vercel.app/
GitHub Repository	https://github.com/eshitaguptaa/SentinelAI-Behavioral-Threat-Detection
Video Demo	https://drive.google.com/drive/folders/1lx4J1kTKugMUaBTTGAO9ncda5uuYa4Xy?usp=sharing



FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

- Deployable using Python and React.
- Supports batch and near real-time predictions.
- Modular architecture for enterprise environments.
- Scalable API-based deployment.

Potential challenges and risks

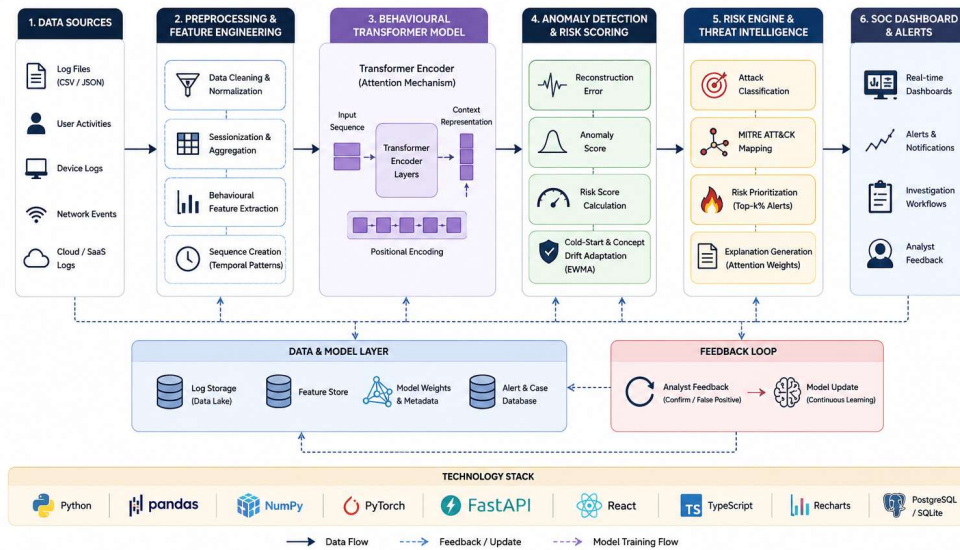
- Class imbalance in cybersecurity datasets.
- Behavioural changes over time (concept drift).
- New users with limited historical data.
- Large-scale enterprise log processing.

Strategies for overcoming these challenges

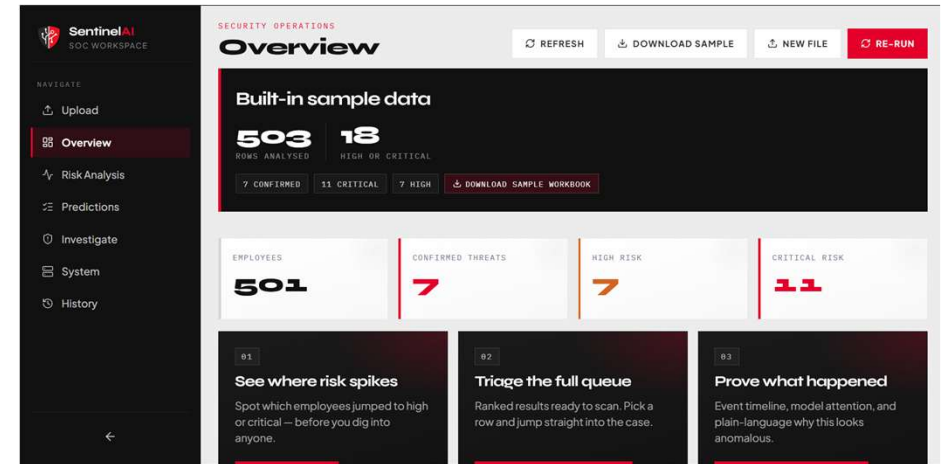
- Behavioural Transformer for unsupervised anomaly detection.
- Cold-start adjustment for new entities.
- Concept drift adaptation using EWMA.
- Explainable AI for analyst confidence.
- Ranked alerts to reduce alert fatigue.



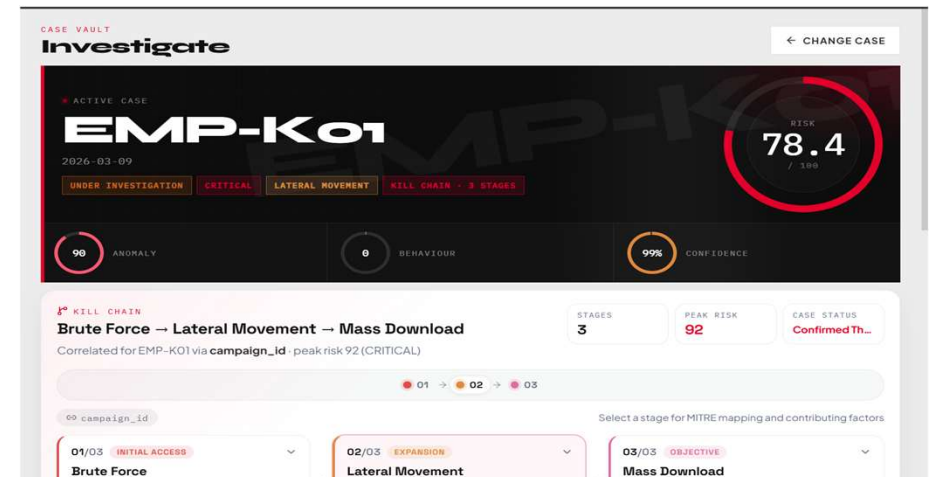
ARTIFACTS



System Architecture



Overview/Dashboard



RESEARCH AND REFERENCES

Research References

- MITRE ATT&CK Framework – <https://attack.mitre.org>
- Attention Is All You Need (Transformer Paper) – <https://arxiv.org/abs/1706.03762>
- PyTorch Documentation – <https://pytorch.org/docs/stable>
- FastAPI Documentation – <https://fastapi.tiangolo.com>
- React Documentation – <https://react.dev>

Project Resources

GitHub Repository: <https://github.com/eshitaguptaa/SentinelAI-Behavioral-Threat-Detection>

Live Demo: <https://sentinel-ai-behavioral-threat-detection-keedprxpu.vercel.app/>

Demonstration:

<https://drive.google.com/drive/folders/1lx4J1kTKugMUaBTTGA09ncda5uuYa4Xy?usp=sharing>

Report : <https://drive.google.com/drive/folders/12JvTOXk1PWGUwuLSoE-WOwRea-8UftAm?usp=sharing>